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The Mutual-Information Paradox in BF16 Weight Streams

Why 2.4 bits of apparent value-space mutual information yield only small bit-level coding gains

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Abstract

We report a quantitative paradox in the inter-value mutual information of BF16 weight streams in trained neural network artefacts. Direct estimation of the conditional entropy $H(v_t | v_{t-1})$ on samples of order 10^3 per context returns $H(v_t | v_{t-1}) \approx 8.1$ bits, suggesting roughly 2.4 bits of mutual information between adjacent BF16 values relative to a value entropy of $H(v) \approx 10.5$ bits. We show that the apparent 2.4 bits is dominated by finite-sample bias of the plug-in entropy estimator, of order $-(K-1)/(2N)$ for K unique values and N samples per context. On a corpus of $\sim 1.8 \times 10^9$ adjacent BF16 pairs we recover the true mutual information at 0.29 bits for exact-value context and 0.004 bits for exponent context. Even the 0.29-bit estimate is diffuse across approxi-

mately 11,000 unique value symbols and is unexploitable at the bit level: context-aware Huffman achieves $1.5212\times$ versus $1.5208\times$ for static Huffman, a relative improvement of 0.03%, and adaptive prediction by partial matching achieves $1.5239\times$, a relative improvement of 0.20%. Bit-level transforms such as exclusive-OR and arithmetic delta of adjacent values increase entropy because the IEEE 754 mapping is non-monotonic in bit space. The result is scoped to public-model BF16 weight tensors at fully-trained checkpoints and is interpreted as evidence that single-model BF16 lossless compression is essentially at its Shannon limit; the apparent value-space correlation is real but trapped in value space and cannot be recovered at the bit level without overhead that exceeds the savings.

Keywords: mutual information; finite-sample bias; trained neural network weights; IEEE 754; BF16; lossless compression; context coding

This is an approved public release of a Forlais Group academic paper.

Methodology Disclosure Boundary

This paper reports the public-safe statement of the FCN mutual-information measurement and the bias correction that resolves it. It does not disclose non-public Forlais implementation details, non-public code, run instructions, source logs, hidden benchmark mechanics, private datasets, prompts, model weights, infrastructure, or operational procedures.

1 Introduction

Lossless compression of trained neural network weights stored in a fixed binary floating-point format is governed by the Shannon entropy of the joint distribution of bits across consecutive stored values [13, 3]. If adjacent stored values are statistically dependent, a context model that conditions on previous values can in principle approach the conditional entropy $H(v_t | v_{t-1})$ rather than the unconditional value entropy $H(v)$.

The FCN evidence register records a measurement that initially appears to support the existence of a substantial inter-value mutual information in BF16 weight streams. Direct estimation of $H(v_t | v_{t-1})$ on tractable per-context sample sizes returns roughly 8.1 bits per value, implying $MI(v_t; v_{t-1}) \approx 2.4$ bits relative to the measured unconditional value entropy of $H(v) \approx 10.5$ bits. If that mutual information were exploitable, the BF16 single-model lossless compression ceiling should sit substantially above the empirically observed $1.52\times$.

It does not. Every context model, prediction model, and bit-level transform we have tested either fails to improve on static entropy coding or pays an overhead that exceeds any conditional savings. We report the resolution of this apparent paradox: the 2.4 bits is dominated by finite-sample bias in the plug-in entropy estimator, the true mutual information is roughly 0.29 bits, and even that smaller quantity is unexploitable at the bit level because the IEEE 754 mapping is non-monotonic and the correlation is diffuse across approximately 11,000 unique value symbols.

The contribution is empirical and methodological. We give the bias correction, the corrected true mutual information, and the documented failure of context, prediction, and transform methods to recover more than a small part of it. The headline measurements are an apparent $MI \approx 2.4$ bits under naive estimation, a corrected $MI = 0.29$ bits under bias-aware estimation on $\sim 1.8 \times 10^9$ adjacent pairs, and directly observed bit-level savings of approximately 0.003 to 0.021 bits per value, depending on coder. This corresponds to roughly 1% to 7% of the corrected value-space mutual information before model-description overhead.

The remainder of the paper states the public-safe scope, summarises the relevant background, presents the verification protocol, gives the theoretical analysis of the bias correction, presents the empirical results across context and prediction methods, explains why bit-level transforms fail, surveys the connection to the public literature, and records the limitations and disclosure boundary.

2 Scope

The result reported in this paper concerns:

- adjacent-value mutual information in BF16 weight streams of trained neural network artefacts;
- public model artefacts at fully-trained checkpoints;
- the single-model single-format lossless setting; and

- the gap between value-space and bit-space mutual information.

The result does not concern:

- lossy regimes such as quantisation-aware training, distillation, or pruning, which alter the distribution under study;
- cross-model delta regimes, where adjacent-value statistics are modified by the conservatism of fine-tuning;
- non-adjacent contexts such as long-range autocorrelation, which are not covered by the immediate-predecessor estimator;
- formats other than BF16 except where explicit comparison is drawn; and
- architectures that do not store weights in a standard IEEE 754 binary format.

The numerical results should not be applied to data outside the tested setting without separate justification.

3 Background and related work

3.1 Shannon entropy and mutual information

For a discrete random variable X taking values in a finite alphabet $\mathcal{X}$, the Shannon entropy is $H(X) = -\sum_x P(x) \log_2 P(x)$, and for two random variables X, Y , the mutual information is $\text{MI}(X; Y) = H(X) + H(Y) - H(X, Y) = H(X) - H(X | Y)$ [13, 3]. A lossless code conditioned on Y can achieve expected length per symbol arbitrarily close to $H(X | Y)$, so a non-zero $\text{MI}(X; Y)$ is in principle exploitable as compression savings of up to $\text{MI}(X; Y)$ bits per symbol relative to a code that ignores Y .

3.2 Finite-sample bias in plug-in entropy estimators

The plug-in estimator $\hat{H} = -\sum_x \hat{P}(x) \log_2 \hat{P}(x)$ that substitutes empirical frequencies for true probabilities is known to be downward biased [9, 10]. To leading order, the bias is

$$E[\hat{H}] - H \approx -\frac{K - 1}{2N \ln 2}, \quad (1)$$

where K is the number of distinct symbols and N is the sample size. For conditional entropy estimated by partitioning a corpus into C contexts of N/C samples each, the per-context bias is correspondingly larger. The Miller-Madow correction adds back $(K - 1)/(2N \ln 2)$ as a first-order fix; higher-order corrections are surveyed in Paninski [10].

3.3 Context coding and prediction by partial matching

Adaptive arithmetic coding under a Markov context model achieves a code length per symbol that approaches the conditional entropy under the modelled context [11, 14, 12]. Prediction by partial matching (PPM) [2] mixes context-coding probabilities across context orders. The Krichevsky-Trofimov estimator [7] provides minimax-optimal universal coding for finite-alphabet sources. None of these methods can in general recover more compression than the true mutual information of the source.

3.4 Trained-weight statistics

Adjacent FCN entries report that trained-weight values within a tensor approximately satisfy IID statistics at the value level, that the mantissa field is Benford-compliant with bit-level biases captured by the $r = 2$ law, and that the BF16 single-model lossless compression ceiling sits near $1.52\times$. The present paper studies the apparent contradiction between an IID claim and a measured MI ≈ 2.4 bits.

4 Verification protocol

We measure unconditional and conditional entropies on BF16 weight streams of public-model artefacts. Unconditional entropy is estimated as $\hat{H}(v) = -\sum_v \hat{P}(v) \log_2 \hat{P}(v)$ over the full stream. Conditional entropy is estimated under two context definitions: exact-value context, in which $P(v_t \mid v_{t-1})$ is conditioned on the full sixteen-bit predecessor; and exponent context, in which the predecessor is reduced to its eight-bit exponent.

For each context definition, we report (a) the naive plug-in estimate at $\sim 10^3$ samples per context, which is the regime where the apparent paradox arises; (b) the Miller-Madow-corrected estimate [9]; and (c) the bias-aware estimate on a much larger corpus ($\sim 1.8 \times 10^9$ adjacent pairs) where finite-sample bias is negligible relative to the quantities of interest. We additionally measure the achievable lossless compression ratio under static Huffman coding [6], context-aware Huffman, and adaptive prediction by partial matching [2], with results expressed as the relative improvement over the static Huffman baseline.

We also report the entropy of two bit-level transforms of the adjacent-value stream: the exclusive-OR $v_t \oplus v_{t-1}$ and the integer delta $v_t - v_{t-1}$ (interpreted as unsigned sixteen-bit subtraction). These transforms expose the gap between value-space and bit-space correlation.

5 Theoretical analysis

5.1 The bias term that produces the apparent 2.4 bits

Let $\mathcal{X}$ denote the alphabet of BF16 values present in the weight stream, with $|\mathcal{X}| = K \approx 11,000$ in the public-model corpora we examine. Partitioning the corpus into context classes by predecessor value gives $C \leq K$ contexts with sample sizes $N_c \in [10^3, 10^4]$ per context in the regime of the original measurement. The Miller bias of the per-context plug-in entropy estimator is

$$E[\hat{H}(v_t \mid v_{t-1} = c)] - H(v_t \mid v_{t-1} = c) \approx -\frac{K_c - 1}{2N_c \ln 2}, \quad (2)$$

where $K_c \leq K$ is the number of distinct v_t symbols observed in context c . The expectation of the conditional entropy estimator is therefore biased downwards by approximately

$$\sum_c P(c) \cdot \frac{K_c - 1}{2N_c \ln 2}. \quad (3)$$

For $K \approx 11,000$ unique values and per-context sample sizes of order 10^3 , this bias is of order 5 to 6 bits in absolute magnitude. The plug-in conditional entropy estimator is therefore biased downward relative to the true conditional entropy by an amount comparable to or larger than the apparent mutual information of 2.4 bits. In other words, the apparent reduction from $H(v) \approx 10.5$ bits to $H(v_t \mid v_{t-1}) \approx 8.1$ bits is dominated by bias rather than by genuine signal.

5.2 The bias-corrected mutual information

On the bias-aware estimate at $\sim 1.8 \times 10^9$ adjacent pairs, the corrected conditional entropies are $H(v_t \mid v_{t-1} = c) \approx 10.21$ bits for exact-value context and $H(v_t \mid v_{t-1} = e) \approx 10.496$ bits for exponent context, giving corrected mutual informations of 0.29 bits and 0.004 bits respectively. The exact-value mutual information of 0.29 bits is real but small. The exponent mutual information is negligible at the precision of the measurement.

5.3 Why 0.29 bits is diffuse and unexploitable

The corrected 0.29 bits is diffuse across approximately 11,000 unique value symbols. For a context coder to exploit Δ bits per symbol on average, it must encode a context-dependent probability distribution that differs from the unconditional distribution by at least the corresponding Kullback-Leibler divergence. The cost of transmitting that conditional distribution is bounded below by the model description length, which for K -symbol alphabets at twenty-four bits per code-length entry and $C \sim K$ contexts is many megabytes per tensor. For typical tensor sizes, this overhead is larger than the savings, and the achievable compression after model description is, at best, marginally above the static Huffman baseline.

5.4 Bit-level non-monotonicity at exponent boundaries

The IEEE 754 representation is structured such that the integer mantissa overflows into the exponent field at the boundary of each binade. For two values $v_1 < v_2$ with $v_1 < 2^{e+1} \leq v_2$, the bit representation jumps from $(e, m_{\max})$ to $(e + 1, 0)$, flipping the most-significant mantissa bit and the least-significant exponent bit. The Hamming distance between these two stored values is at least two and is typically much larger when v_2/v_1 is close to unity. The exclusive-OR of v_1 and v_2 therefore has a high-entropy bit pattern despite the small value-space delta.

This non-monotonicity is a general property of binary floating-point representations and is the precise mechanism by which value-space correlation is masked in bit space. Any bit-level codec that operates on adjacent stored values without first inverting the floating-point mapping will encounter this masking and will fail to exploit the underlying correlation.

5.5 A direct upper bound on bit-level exploitable mutual information

We can give a direct upper bound on the bit-level exploitable mutual information $\text{MI}_{\text{bit}}(v_t; v_{t-1})$ as follows. Let $\phi : \mathcal{X} \rightarrow \{0, 1\}^{16}$ denote the IEEE 754 mapping from BF16 values to sixteen-bit codes, and let ϕ^{-1} denote its inverse. The bit-level mutual information of the stream is at most the value-level mutual information, because any bit-level transform can be implemented as a value-level transform followed by encoding, and the data processing inequality [3] bounds the bit-level mutual information from above by the value-level mutual information. With $\text{MI}_{\text{value}}(v_t; v_{t-1}) = 0.29$ bits, we have $\text{MI}_{\text{bit}}(v_t; v_{t-1}) \leq 0.29$ bits. The measured bit-level savings of approximately 0.003 to 0.021 bits per value are therefore consistent with coders that capture only a small fraction of the value-level mutual information before description-length and adaptation costs absorb the rest.

6 Empirical results

6.1 Mutual-information estimates

Table 1 reports the four mutual-information estimates under the three estimation regimes.

Context	Regime	Sample size	MI (bits)
Exact value	Naive plug-in	$\sim 10^3$ per context	2.40
Exact value	Miller-Madow	$\sim 10^3$ per context	0.34
Exact value	Bias-aware	$\sim 1.8 \times 10^9$ pairs	0.29
Exponent	Bias-aware	$\sim 1.8 \times 10^9$ pairs	0.004

Table 1: Mutual-information estimates between adjacent BF16 values under three estimation regimes for two context definitions. The naive plug-in estimate is dominated by finite-sample bias; the Miller-Madow first-order correction reduces the estimate by an order of magnitude; the bias-aware estimate at a large corpus stabilises at 0.29 bits for exact-value context and at 0.004 bits for exponent context.

6.2 Compression methods

Table 2 reports the achievable lossless compression ratio under four representative methods on a BF16 weight tensor of the public model artefact used in the bias-aware measurement. The relative improvement over the static Huffman baseline is expressed in basis points.

Method	Ratio	vs static Huffman
Static Huffman	1.5208	baseline
Context Huffman (exponent)	1.5212	+0.03%
Adaptive PPM	1.5239	+0.20%
Rank-delta of adjacent values	1.4003	worse

Table 2: Lossless compression ratios under four representative methods on a BF16 weight tensor of a public-model artefact. Static Huffman is the entropy-coding baseline. Context Huffman and adaptive PPM are the two natural ways to exploit the corrected 0.29 bits of mutual information; both remain below 0.25% relative-ratio improvement over the baseline. Rank-delta is a representative bit-level transform; it makes compression strictly worse.

The exploitation rate, defined as the realised improvement divided by the corrected mutual information, is approximately 0.001 for context Huffman and 0.00038 for adaptive PPM. In both cases the savings are within the noise of the measurement.

7 Why bit-level transforms fail

The IEEE 754 BF16 representation is non-monotonic in bit space at exponent boundaries. Two values that are numerically close, for example 0.499 stored as 0x3EFF and 0.500 stored as 0x3F00, differ by nine bit positions despite a value-space delta of approximately 0.001. A bit-level transform that operates on adjacent stored values therefore turns small value-space differences into large Hamming distances at the exponent boundary.

We measured the entropy of two such transforms:

- the exclusive-OR of adjacent values, $v_t \oplus v_{t-1}$, has empirical entropy $H(\oplus) \approx 11.4$ bits per pair, which is above $H(v) \approx 10.5$ bits per value;
- the integer delta of adjacent values, $v_t - v_{t-1}$ as unsigned sixteen-bit subtraction, has empirical entropy $H(\Delta) \approx 11.5$ bits per pair, similarly above $H(v)$.

Both transforms therefore increase rather than decrease the entropy of the stream. The reason is

straightforward: the value-space mutual information of 0.29 bits is invisible to a bit-level transform because the transform is sensitive to bit patterns rather than to numerical proximity. Adjacent stored values that are numerically close but on opposite sides of an exponent boundary contribute high-Hamming-distance pairs to the transformed stream and inflate its bit-level entropy.

This is the precise sense in which the 0.29 bits is trapped in value space. The information is real, but the only encoder that can access it must condition on the full predecessor value at the value-symbol level, paying an associated model description cost that exceeds the savings.

8 Cross-context comparison of mutual-information estimates

8.1 Long-range autocorrelation is also small

We have verified that the inter-value mutual information is small not only at lag one (the adjacent-value estimator reported above) but at lags up to several thousand within a tensor. The empirical autocorrelation function of the BF16 weight stream on a representative public-model tensor is consistent with fair-coin statistics at every tested lag, with $\rho \in [-0.01, 0.01]$ for lags between one and 4096. The mutual information between v_t and v_{t-k} for k in this range is therefore similarly small to the adjacent-value mutual information and is similarly unexploitable at the bit level.

8.2 Per-context sample size dependence

The Miller bias scales as $-(K-1)/(2N \ln 2)$ in the per-context sample size N . To verify that the apparent mutual information of 2.4 bits is indeed an artefact of the per-context sample size rather than of the corpus aggregation, we measured the apparent mutual information at three per-context sample sizes: $N \approx 10^3$ gives apparent MI ≈ 2.4 bits, $N \approx 10^5$ gives apparent MI ≈ 0.7 bits, and $N \approx 10^7$ gives apparent MI ≈ 0.32 bits. The trend is consistent with the predicted bias scaling, and the bias-aware estimate at $\sim 1.8 \times 10^9$ pairs stabilises near 0.29 bits.

8.3 Algebraic randomness corroboration

The FCN evidence register records additional measurements that BF16 weight streams within a tensor are algebraically random in the sense of Berlekamp-Massey linear-complexity profile [1, 8]. The empirical linear complexity over $\text{GF}(2^{16})$ is exactly $n/2$ for stream length n , which is the value attained by an algebraically random sequence. Polynomial fits of degree one through five give zero exact matches, autocorrelation at all tested lags is consistent with fair-coin statistics, and the difference sequence diverges (rather than converging, as it would for a structured signal). The mutual-information result reported here is consistent with these complementary algebraic-randomness measurements: a sequence that is algebraically random at the bit level cannot harbour exploitable mutual information at the bit level.

9 Connection to public literature

9.1 Miller bias and entropy estimation

The Miller bias [9] and its higher-order analogues [10] are standard knowledge in entropy estimation. The novelty of our result is the specific magnitude of the bias for the BF16 weight stream regime: with $K \approx 11,000$ unique values and per-context sample sizes of order 10^3 , the bias is large enough to produce an apparent two-to-three-bit mutual information where the true value is below half a bit.

9.2 Context coding and trained-weight compression

Context-aware entropy coding for neural network weights has been explored in both lossy and lossless regimes [4, 5]. The lossless results, in particular DFloat11 [5], exploit the concentrated exponent field rather than the inter-value mutual information. Our result is consistent with that practice: the achievable lossless savings on BF16 weights come almost entirely from the exponent field, not from inter-value context.

9.3 Algebraic-randomness corroboration

The FCN evidence register records additional corroborating measurements that BF16 weight streams within a tensor are algebraically random at the level of polynomial fits, autocorrelation, and linear-recurrence synthesis [1, 8]. The bit-level autocorrelation is consistent with fair-coin statistics at every lag tested, and the Berlekamp-Massey linear complexity over $\text{GF}(2^{16})$ is exactly $n/2$ for stream length n , which is the value attained by an algebraically random sequence. The mutual-information result reported here is consistent with these complementary measurements.

10 Implications

10.1 IID coding is essentially optimal for single-model BF16

The corrected 0.29 bits of value-space mutual information, of which less than one per cent is recoverable at the bit level, is the quantitative reason that static entropy coding is essentially optimal for single-model BF16 weights. There is no context model, predictor, or transform that can do meaningfully better than the static $1.52\times$ ceiling because the underlying source is, at the level of access available to bit-level codecs, IID.

10.2 The Benford floor and the MI ceiling combine

The adjacent Benford-theory entry establishes that the mantissa entropy is Benford-locked at approximately 6.972 bits per value. The present entry establishes that the residual inter-value structure is below half a bit and is unexploitable at the bit level. Together these two observations give a closed account of why the single-model BF16 lossless compression ceiling is stable across architectures.

10.3 Methodological hygiene for high-alphabet conditional entropy

The bias correction reported here is a particular instance of a more general methodological point. Whenever conditional entropy is estimated by partitioning a sample into context classes, the per-context sample size N_c must substantially exceed the per-context effective alphabet size K_c in order for the plug-in estimator to be reliable. For BF16 weight streams, the practical effective alphabet within any one tensor is of order 10^4 distinct values, while typical per-context sample sizes on a single tensor are of order 10^3 . This ratio inverts the usual regime in which plug-in entropy estimation is unbiased, and a bias correction (Miller-Madow at leading order, NSB or jackknife at higher orders [10]) is required to recover the true conditional entropy. We record this point because the discrepancy between the apparent and true mutual information observed in our setting is large enough to invalidate qualitative conclusions, not only quantitative ones.

10.4 The result limits, but does not eliminate, future single-model work

The result shows that no current context coder operating at the bit level can exceed the BF16 ceiling by more than measurement noise on the public-model artefacts examined. It does not rule out the existence of a coder whose model description cost is asymptotically smaller than the naive context table. Methods such as universal source coding under the Krichevsky-Trofimov estimator [7], context tree weighting, and asymptotic Bayesian mixture coding could in principle achieve a description-length cost that scales sub-linearly in the alphabet size. We do not in this paper rule out such an approach; we report that the methods we have tested do not achieve material savings.

10.5 The bias correction is a methodological warning

The naive plug-in estimator gives an apparent mutual information that is wrong by approximately one order of magnitude. We note this explicitly because the magnitude of the bias is large enough to be material to engineering and research decisions about compression-method selection. Any compression project that estimates mutual information on per-context sample sizes of order 10^3 on alphabets of order 10^4 should apply the Miller-Madow correction or work at sample sizes several orders of magnitude larger.

11 Implications for codec selection

11.1 Static codes are sufficient for single-model BF16

The corrected 0.29 bits of value-space mutual information, only a small fraction of which is recoverable at the bit level, means that static Huffman or arithmetic coding is essentially optimal for single-model BF16 weight streams. The marginal benefit of context modelling, prediction by partial matching, or grammar-based coding is below the noise of the measurement on the public-model artefacts examined. A storage system that selects a codec for single-model BF16 weight tensors should default to static near-entropy coding and should not invest in context models.

11.2 Context modelling is worth attempting only outside the single-model regime

The cross-model delta regime documented in the adjacent Fine-Tuning Conservatism Law entry does admit substantial savings beyond the single-model ceiling. In that regime, the residual mantissa is concentrated near zero and the value-level statistics are no longer Benford-locked, so context-aware coding can exploit the per-column structure noted in the adjacent paper. Outside the cross-model delta regime, however, context modelling is essentially zero-improvement for the BF16 streams we have examined.

11.3 The bias correction as a default

Whenever conditional entropy is to be estimated on a BF16 weight stream, the Miller-Madow correction should be applied as a default. The magnitude of the bias is large enough on per-context sample sizes of order 10^3 to invalidate qualitative conclusions about exploitability, and the correction is computationally trivial. The FCN evidence register applies this correction throughout, and the corrected estimates form the basis of the ceiling analysis in the adjacent Benford-theory entry.

12 Discussion

The mutual-information paradox is resolved as follows. The apparent 2.4 bits of mutual information between adjacent BF16 weight values is dominated by finite-sample bias of the plug-in conditional entropy estimator. The corrected mutual information is 0.29 bits, smaller than the unconditional value entropy by a factor of approximately thirty-five. Even the corrected 0.29 bits is diffuse across approximately 11,000 unique value symbols, so any context model that attempts to exploit it pays a description-length cost that exceeds the achievable savings. Bit-level transforms such as exclusive-OR and integer delta increase rather than decrease the entropy of the stream because the IEEE 754 mapping is non-monotonic in bit space.

The result is a quantitative complement to the Benford derivation of the BF16 ceiling. Benford explains why the mantissa entropy sits at 6.972 bits per value. The mutual-information measurement explains why the residual inter-value structure does not move the ceiling: the structure exists in value space but not in bit space, and the value-space structure is too diffuse to support a context coder that is net positive after model description costs.

We do not claim that no method can ever exceed the ceiling. The cross-model delta regime documented in the adjacent Fine-Tuning Conservatism Law entry exceeds it consistently. We do claim that no single-model single-format method that operates at the bit level on the value stream we examined can exceed the ceiling by more than the noise of the measurement.

13 The result in the broader FCN context

13.1 Position within Category 22

The MI Paradox is one of five empirical regularities documented in the FCN Category 22 fundamental-law programme. The other four are the $r = 2$ law (mantissa-bit bias geometric progression), the Benford-theory derivation (closed-form mantissa entropy), the FP32 ceiling (format-width analogue of the BF16 ceiling), and the Fine-Tuning Conservatism Law (cross-model delta exception). The unified mathematical account of the BF16 ceiling, presented in the adjacent Math Insights entry, integrates all five regularities into a single picture.

The MI Paradox plays a particular role in that picture. It demonstrates that the residual within-tensor mutual information, which is the natural candidate for any extension of single-model compression beyond the ceiling, is too small to support a context coder that is net positive after model description costs. Without this demonstration, one might reasonably hypothesise that a sufficiently clever context coder could exceed the ceiling by exploiting the apparent 2.4 bits of inter-value mutual information. The bias correction rules out this hypothesis.

13.2 Connection to algebraic-randomness measurements

The MI Paradox is consistent with the FCN finding that BF16 weight streams within a tensor are algebraically random in the sense of Berlekamp-Massey linear-complexity profile [1, 8]. Both measurements indicate that the bit-level structure of the stream is consistent with fair-coin statistics modulo the Benford-locked mantissa biases. The MI Paradox provides the quantitative bound on the residual structure; the algebraic-randomness measurements provide an independent corroboration.

13.3 Relation to the bit-versus-value gap

The bit-versus-value gap, documented in the present paper and in the adjacent Math Insights entry, is the structural property that makes the value-space mutual information invisible to bit-level codecs. Under the IEEE 754 representation, two numerically close values can sit at large Hamming distance because of the non-monotonic mapping at exponent boundaries. The bit-level exclusive-OR of adjacent values inherits this non-monotonicity and is statistically indistinguishable from a random bit stream. The MI Paradox is therefore not a peculiarity of any particular coder; it is a structural consequence of the IEEE 754 representation that bounds bit-level coders.

14 Limitations

We note five limitations.

1. The mutual-information estimates are for the adjacent-pair context only. Long-range autocorrelation is not addressed in this paper; the FCN evidence register records that autocorrelation at lags up to 4096 is consistent with fair-coin statistics, but a separate paper would be needed to make the long-range claim formally.
2. The bias-corrected estimate of 0.29 bits is itself an estimate. We do not claim it is exact; we claim it is consistent across the public-model artefacts examined and is small enough that the exploitation question is settled in the negative for bit-level codecs.
3. The achievable compression numbers in Table 2 are method-specific. A novel context model with a substantially smaller description-length cost could in principle exploit more of the 0.29 bits, but we have not found one. We make this distinction explicit because exhaustive impossibility is not what we claim.
4. The result is for BF16. The corresponding measurements for FP32 and FP16 are recorded separately and are consistent with the BF16 picture; the present paper does not extend the formal claim to those formats.
5. Public reproducibility is constrained by disclosure control. Internal verification artefacts that produced the measurements are maintained by Forlais Group but are not included in this paper.

15 Data and code availability

Per-experiment records, verification artefacts, and raw measurements are maintained by Forlais Group. Materials can be made available on reasonable request subject to disclosure review, confidentiality restrictions, and removal of proprietary methodology or operational detail. This paper does not promise public release of any internal artefact beyond the abstraction level reported here. The public-model artefacts used in the measurements are released by their respective developers under their stated licences.

16 Competing interests

The author is affiliated with Forlais Group Ltd, which conducts the FCN compression research programme described in this paper. The author declares no other competing financial or non-financial interests.

17 Conclusion

We report and resolve the apparent mutual-information paradox in BF16 weight streams. The naive plug-in estimator returns $MI \approx 2.4$ bits between adjacent BF16 values; the bias correction reduces this to 0.29 bits at exact-value context and to 0.004 bits at exponent context on a corpus of $\sim 1.8 \times 10^9$ adjacent pairs. The residual mutual information is diffuse and unexploitable at the bit level: context-aware Huffman improves over static Huffman by 0.03%, adaptive PPM improves by 0.20%, and bit-level transforms (exclusive-OR, integer delta) strictly increase entropy. The IEEE 754 mapping is non-monotonic in bit space, so value-space correlation is invisible to bit-level codecs. The result is a quantitative explanation of why static entropy coding is essentially optimal for single-model BF16 weights and complements the Benford-derived account of the $1.52\times$ compression ceiling. The result is reported in its public-safe form and should be read alongside the limitations enumerated above.

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Revision History

Version	Date	Change summary	Owner
0.1	14 May 2026	Initial FCN academic paper package generated from source evidence.	Oliver Cooper
1.0	14 May 2026	Full evidence-grounded manuscript with bias analysis, corrected mutual-information estimates, context-coding and bit-level-transform results, and connection to public literature on finite-sample entropy estimation.	Oliver Cooper